DG INTERN HUB

WEEK 4

CYBER SECURITY INTERNSHIP TASK

Basic Network Setup • Web Server • Nmap • Wireshark

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Organization	DG Intern Hub
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Lab target	Windows host on VirtualBox Host-Only network

Practical evidence report with screenshots captured during the lab. Findings are recorded as observed; unsuccessful DNS resolution is documented transparently.

1. Objective and Lab Overview

The goal of Week 4 was to build a small lab network, bring up a web service, scan the target with Nmap, and capture application/network traffic in Wireshark. The work was performed in an authorized local lab using a Kali Linux VM and the Windows host connected through VirtualBox Host-Only networking.

Lab addressing

Node	Role	Address / Interface
Kali Linux VM	Test client / traffic capture	192.168.56.101/24 on eth1
Windows host	Target web server (IIS)	192.168.56.1/24 — Ethernet 2
Host-Only network	Isolated lab segment	192.168.56.0/24

Workflow

1) Verify Host-Only addresses → 2) Start and test Apache on Kali → 3) Enable IIS on Windows → 4) Run initial and final Nmap scans → 5) Capture HTTP and ICMP traffic in Wireshark → 6) Record DNS resolution limitation.

Scope note: this was a small two-node lab (Kali VM plus Windows host). No additional third VM or FTP service was configured.

2. Network Configuration Evidence

The Host-Only network was used to keep the practical traffic on the lab segment. Kali's eth1 interface received 192.168.56.101/24 and the Windows Host-Only adapter was 192.168.56.1/24.

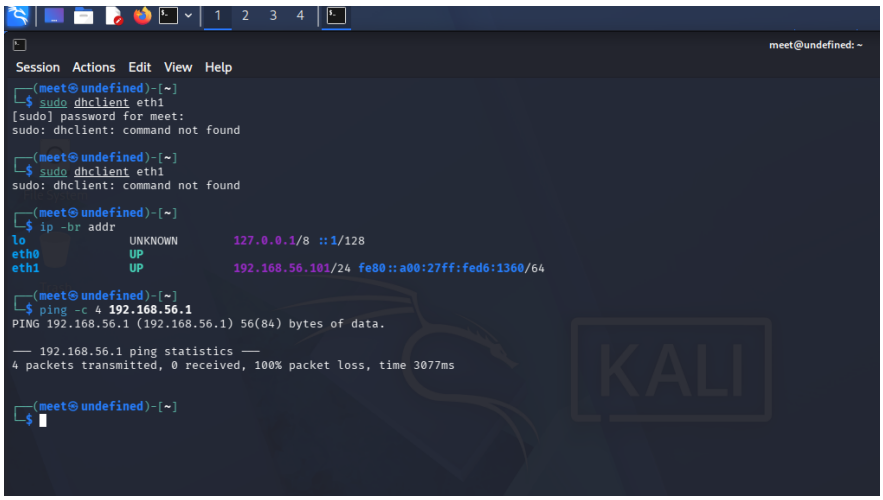


Figure 1. Kali network interface and Host-Only address; ping replies were not received during this check.

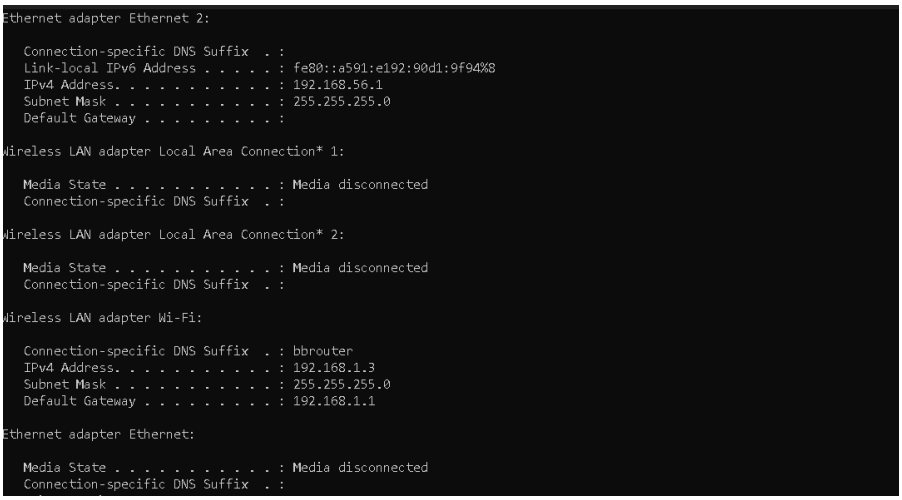


Figure 2. Windows Ethernet 2 Host-Only adapter showing IPv4 192.168.56.1 and subnet mask 255.255.255.0.

3. Web Server Setup — Apache on Kali

Apache2 was already installed on Kali. The service was started and verified as active. The default Debian Apache page loaded in the browser, the server listened on TCP port 80, and curl returned HTTP/1.1 200 OK. This validates the local Apache installation; it is separate from the Windows IIS target used for the Nmap scan.

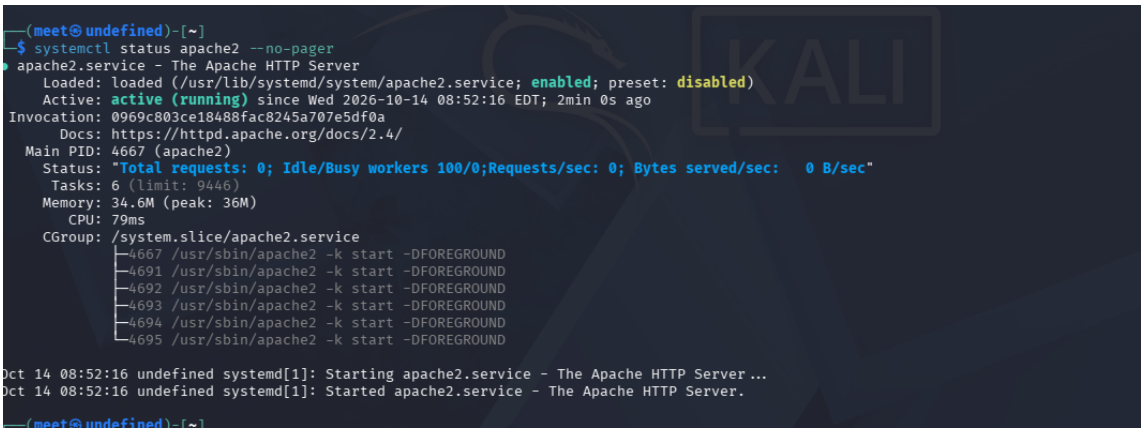


Figure 3. Apache2 service status shows active (running).

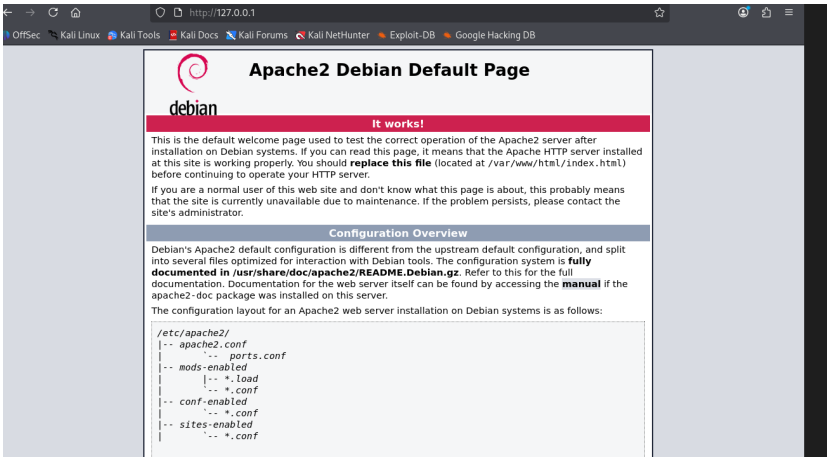


Figure 4. Apache2 Debian default page displayed at http://127.0.0.1.

4. Web Server Verification — HTTP and IIS

The local Apache listener was confirmed on port 80 and a HEAD request returned HTTP 200 OK. IIS was then enabled on the Windows host for the target-side web service. The Windows browser displayed the IIS welcome page.

```
┌──(meet㉿undefined)-[~]
└─$ sudo ss -ltnp | grep ':80'
LISTEN 0      511                *:80              *:*    users:(("apache2",pid=8295,fd=4),("apache2",pid=4695,fd=4),("apache2",pid=4694,fd=4),("apache2",pid=4693,fd=4),("apache2",pid=4692,fd=4),("apache2",pid=4691,fd=4),("apache2",pid=4667,fd=4))
```

Figure 5. Apache2 listening on TCP port 80.

```
┌──(meet㉿undefined)-[~]
└─$ curl -I http://127.0.0.1
HTTP/1.1 200 OK
Date: Wed, 14 Oct 2026 13:00:53 GMT
Server: Apache/2.4.68 (Debian)
Last-Modified: Wed, 01 Jul 2026 05:30:38 GMT
ETag: "29cf-65585fde3f36c"
Accept-Ranges: bytes
Content-Length: 10703
Vary: Accept-Encoding
Content-Type: text/html
```

Figure 6. curl -I returned HTTP/1.1 200 OK from the local Apache service.

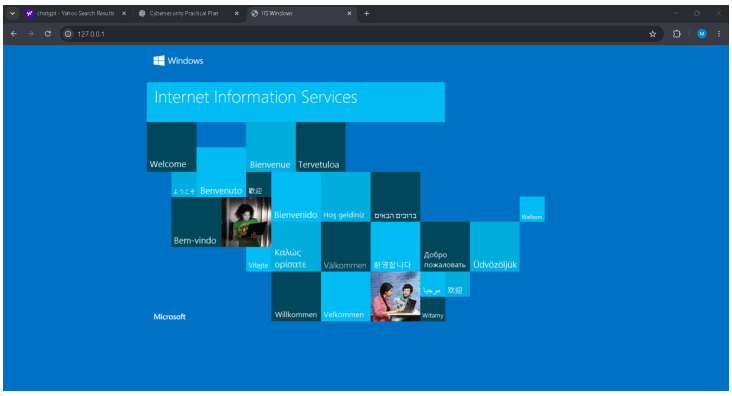


Figure 7. Windows IIS welcome page at http://127.0.0.1.

5. Nmap — Initial Target Scan

The first targeted scan checked TCP ports 80, 3306 and 7070 on 192.168.56.1. At this point, port 80 was filtered, while MySQL on 3306 and the service detected as ssl/realserver? on 7070 were reported open. This motivated enabling the Windows IIS web role and verifying port 80 again.

Command used

sudo nmap -Pn -sV -p 80,3306,7070 192.168.56.1 -oN ~/week4-initial-scan.txt

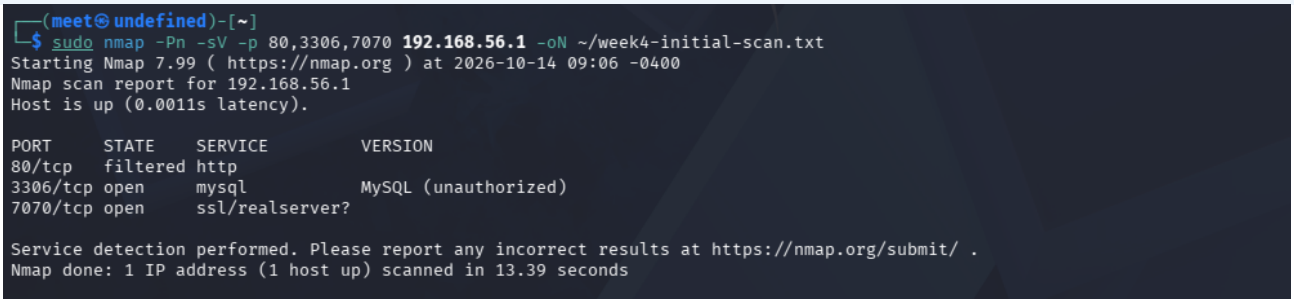


Figure 8. Initial Nmap scan: 80/tcp filtered; 3306/tcp open mysql; 7070/tcp open ssl/realserver?.

6. Windows IIS Enablement and Port 80 Verification

The Windows IIS feature was enabled. The first Nmap recheck showed TCP port 80 open, confirming that the web service was reachable from Kali over the Host-Only network. The final service scan identified the service as Microsoft IIS HTTPD 10.0.

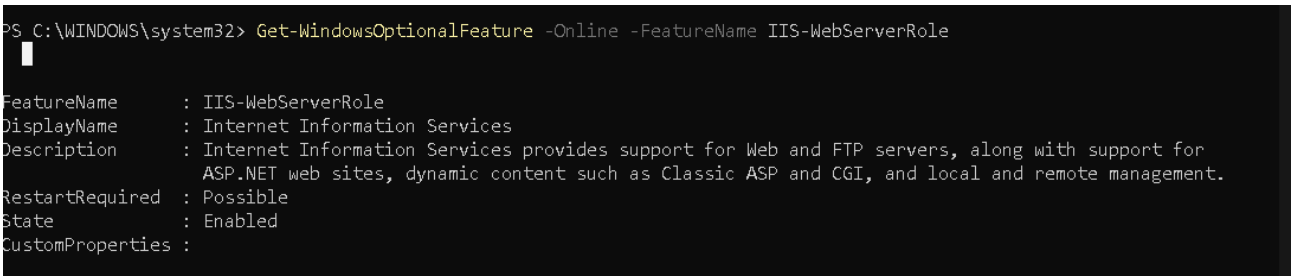


Figure 9. Windows feature query shows IIS-WebServerRole enabled; restart was listed as possible.

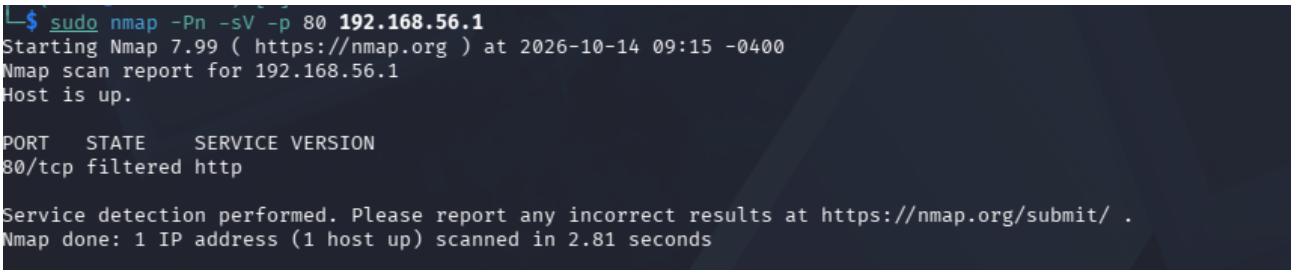


Figure 10. Nmap verification shows 80/tcp open http.

7. Nmap — Final Service Scan

The final scan recorded TCP port 80 as open and identified Microsoft IIS HTTPD 10.0. Ports 3306 and 7070 were filtered in this final check. Nmap output was saved to ~/week4-final-scan.txt and displayed for evidence.

Command used

sudo nmap -Pn -sV -p 80,3306,7070 192.168.56.1 -oN ~/week4-final-scan.txt



Figure 11. Final Nmap result: 80/tcp open (Microsoft IIS HTTPD 10.0); 3306/tcp and 7070/tcp filtered.

8. Wireshark — HTTP Traffic Analysis

Wireshark captured traffic on Kali's eth1 interface with the display filter http. The capture shows a GET / request from 192.168.56.101 to 192.168.56.1 and an HTTP/1.1 200 OK response. Additional requests for image and favicon resources were also observed; one resource returned 404 Not Found.

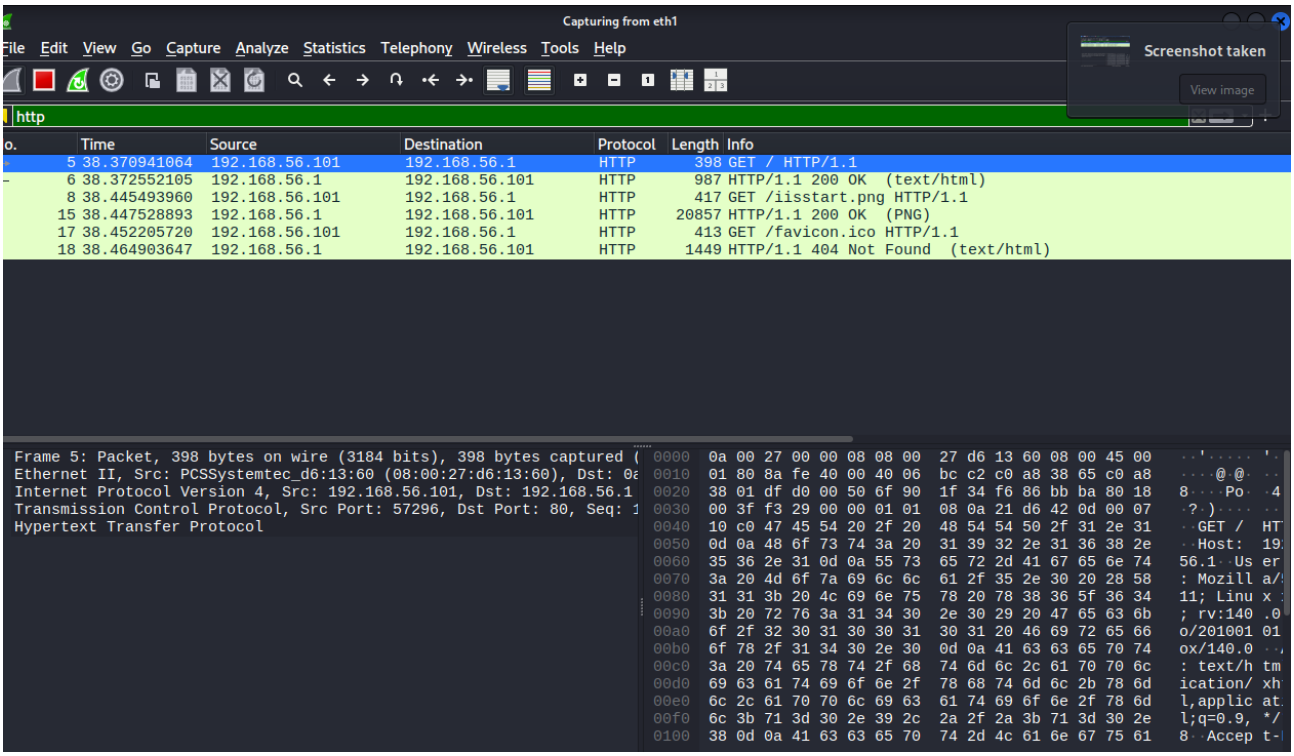


Figure 12. HTTP GET and 200 OK response visible in Wireshark on eth1.

Interpretation: the HTTP exchange confirms application-layer connectivity from Kali to the Windows IIS target over the lab network.

9. Wireshark — ICMP and DNS Findings

ICMP filter results show four Echo Request packets from Kali (192.168.56.101) to Windows (192.168.56.1), with no Echo Reply packets visible in the capture. This is consistent with ICMP replies being blocked or not returned; the capture alone does not prove the exact cause.

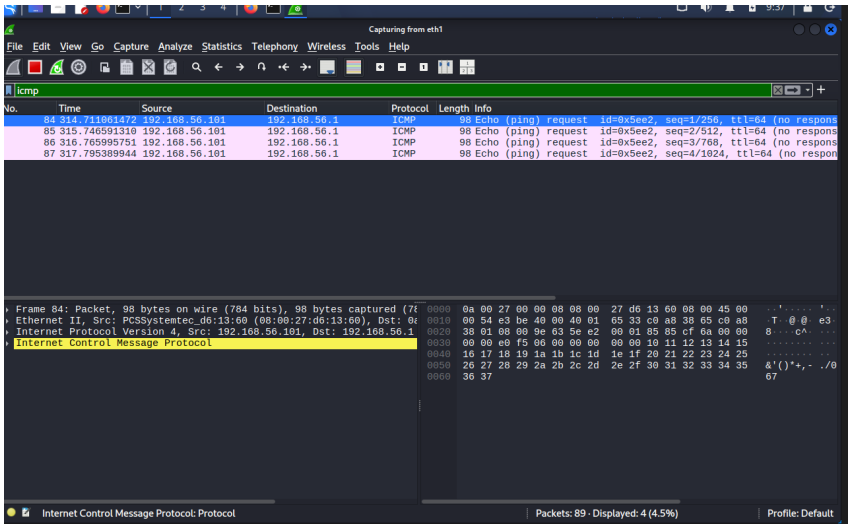


Figure 13. ICMP Echo Requests captured; no replies are shown.

DNS attempt

The nslookup example.com command failed with “network unreachable” while trying DNS server 192.168.1.1. Therefore, successful DNS query/response traffic was not captured in this run. This is recorded as a limitation rather than a successful DNS finding.

```
┌──(meet㉿undefined)-[~]
└─$ nslookup example.com
;; UDP setup with 192.168.1.1#53(192.168.1.1) for example.com failed: network unreachable.
;; no servers could be reached
;; UDP setup with 192.168.1.1#53(192.168.1.1) for example.com failed: network unreachable.
;; no servers could be reached
;; UDP setup with 192.168.1.1#53(192.168.1.1) for example.com failed: network unreachable.
;; no servers could be reached
```

Figure 14. DNS lookup attempt failed because the configured resolver was unreachable.

10. Findings, Limitations and Conclusion

Key findings

Area	Observed result	Interpretation
Network	Kali 192.168.56.101/24; Windows 192.168.56.1/24	Host-Only lab segment established.
Web server	Windows TCP/80 open; Microsoft IIS HTTPD 10.0	Target web service reachable from Kali.
Other ports	3306 and 7070 filtered in final scan	Nmap did not receive normal responses; filtered is not the same as closed.
HTTP	GET / and HTTP/1.1 200 OK captured	HTTP request/response exchange verified.
ICMP	Four Echo Requests; no replies shown	Reachability via ICMP was not confirmed.
DNS	nslookup failed: network unreachable	Successful DNS capture remains incomplete.

Limitations

DNS resolution could not be demonstrated because the configured DNS server was unreachable from the Kali environment. ICMP replies were not observed. The lab used two nodes rather than three, and no FTP service was configured. These points should be stated accurately if the internship rubric expects all optional/bonus items.

Conclusion

The practical established a Host-Only test segment, verified web service operation, identified the Windows IIS service using Nmap, and captured HTTP application traffic in Wireshark. The work demonstrates a basic workflow for network setup, service discovery and packet analysis. The remaining improvement is to repair DNS connectivity and capture a successful DNS exchange if time permits.